David M. Henke

Computational Biologist

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Summary

Computational biologist and biostatistician with nearly two decades of experience advancing public health and biomedical research in viral and human genetics. Proven success leading interdisciplinary collaborations, guiding research design, mentoring team members, and delivering impactful results through statistical modeling, machine learning, and bioinformatics.

Experience

Baylor College of Medicine, Senior Biostatistician — Houston, TX
Dec 2017 – present
8 years 11 months

- Lead data-driven arm of clinical and basic science research in viral genomics and clinical biomarker discovery
- Direct cross-functional teams of clinicians, computational biologists, and epidemiologists across multiple NIH- and industry-sponsored projects
- Developed BioPrimeLASSO — a network-informed LASSO framework applied to 453 cancer targets across 1,048 DepMap cell lines — published in npj Precision Oncology (2025)
- Spearheaded SARS-CoV-2 pandemic response programs balancing data urgency with scientific rigor
- Peer reviewer for npj; advise students and collaborators on data-based research design
- Co-lead grant and manuscript submissions across virology, oncology, and genomics

Baylor College of Medicine, Statistician — Houston, TX
July 2016 – Nov 2017
1 year 5 months

- Managed multiple simultaneous projects resulting in several peer-reviewed publications
- Applied advanced non-linear regression models to reveal associations between genetic variables and clinical outcomes in longitudinal and cross-sectional data

Baylor College of Medicine, Bioinformatician — Houston, TX
Mar 2015 – June 2016
1 year 4 months

- Developed bioinformatic pipelines to process and analyze terabytes of next-generation sequencing datasets
- Engineered algorithm enabling clinical non-invasive prenatal testing (cfNIPT)
- Leveraged high-performance computing to align millions of sequencing reads to the human genome
- Built interactive web applications to support collaborative data analysis (R Shiny)

Independent Statistical Consultant, Statistical Advisor — Chapel Hill, NC · Houston, TX · Remote
Apr 2013 – present
13 years 7 months

- Clients include Renovion, Baylor College of Medicine, UNC-Chapel Hill, and PGENI
- Contributed investor deck analysis and preliminary clinical data visualization that secured orphan drug designation for an early-stage pharmaceutical company
- Designed study protocols, analysis plans, IRB forms, and clinical research instruments
- Analyzed data for sample size, interim statistics, and exploratory outcome associations across pharmacological and medical device studies

Education

University of North Carolina Gillings School of Global Public Health, Biostatistics — Chapel Hill, NC
2012 – 2014

- Thesis: Logistic Regression Modeling — Compensating for an Incomplete Covariate via Use of Weights

University of Colorado Boulder, Biochemistry (Minor: Chemistry) — Boulder, CO
2004 – 2008

Skills

Statistical & Machine Learning

Bioinformatics & Genomics

Programming & Tools

Interactive Web & Visualization

Leadership & Communication

Projects

BioPrimeLASSO
R package incorporating protein–protein interaction network scores from STRING DB into the LASSO regularization penalty. Applied to 453 cancer targets across 1,048 DepMap cell lines; recovered GAB2 as an EGFR dependency biomarker.

- Published in npj Precision Oncology (2025)
- doi: 10.1038/s41698-025-00825-9

Preprint Digest
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Individual LLM Deployment
Framework for deploying open-source LLMs on personal or institutional hardware.

LLM Google Colab Deployment
Ready-to-run notebooks for GPU-accelerated LLM inference in Google Colab.

Certificates

Innovation & Business Strategy 2019

Interests

Outside the Lab